

Spesifikasi Teknis Dataset LC dan PS

Dataset ini merupakan kumpulan data eksperimental deteksi logam berat Cadmium menggunakan perangkat **LC dan PS**. Dataset ini disusun untuk menunjukkan stabilitas, repetibilitas, dan akurasi perangkat

1. Hierarki dan Volume Data

Data dikumpulkan secara sistematis dengan pengulangan tinggi untuk memastikan validitas statistik:

- **Total Sampel:** Dataset mencakup total **300 file pengukuran** (15 folder variasi x 50file csv x 10 pengambilan berdasarkan variasi konsentrasi).
- **Variasi Konsentrasi (PPM):** Terdapat 15 tingkatan konsentrasi Cadmium (2, 4, 6, 8, 10, 20, 40, 80, 100, 200, 400, 600, 800 ,1000).
- **Repetisi (Replicate):** Setiap variasi konsentrasi dilakukan pengukuran sebanyak **50 kali pengukuran** untuk menguji presisi alat.
- **Siklus per Pengukuran (Scan):** Setiap satu file pengukuran (1 kali jalan) terdiri dari **10 scan/cycle** *Cyclic Voltammetry* berkelanjutan.

2. Struktur Internal Data Per File

Setiap file .csv/.xlsx hasil ekspor dari Alat berisi baris data yang mencakup 10 siklus voltammetri lengkap. Data disusun secara kronologis:

- **Scan 1-10:** Digunakan untuk melihat stabilitas elektrode. (Biasanya scan ke-3 hingga ke-10 digunakan untuk analisis karena kondisi permukaan elektrode sudah stabil).
- **Data Point:** Setiap scan mengandung sekitar 400-800 titik data (tergantung pada *Scan Rate* dan *Step Potential* yang diatur)

Voltase dan Ampere

AutoSave Off Cad_EM_100ppm_02.xlsx

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Chart Design Format

Clipboard Font Alignment Number Styles

Chart 1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	CV i vs E Scan	CV i vs E Scan 2		CV i vs E Scan 3		CV i vs E Scan 4	CV i vs E Scan 5		CV i vs E Scan 6	CV i vs E Scan 7		CV i vs E Scan 8	CV i vs E Scan 9		CV i vs E Scan 10					
2	V	μA	V	μA	V	μA	V	μA	V	μA	V	μA	V	μA	V	μA	V	μA	V	μA
3	-0,49999	-9,95371	-0,49999	27,80745	-0,49999	30,71971	-0,49999	31,81886	-0,49999	32,86164	-0,49999	32,44829	-0,49999	32,72073	-0,49999	33,07771	-0,49999	32,69254	-0,49999	33,24681
4	-0,51	-9,00574	-0,51	22,57476	-0,51	26,01312	-0,51	26,99014	-0,51	27,69472	-0,51	27,63835	-0,51	29,00993	-0,51	28,51203	-0,51	27,79805	-0,51	28,12686
5	-0,52	-9,19533	-0,52	19,28672	-0,52	23,0257	-0,52	23,32632	-0,52	24,0215	-0,52	23,97453	-0,52	25,05489	-0,52	24,867	-0,52	24,16242	-0,52	24,80124
6	-0,53001	-8,86465	-0,53001	16,60931	-0,53001	20,03827	-0,53001	20,41405	-0,53001	21,45683	-0,53001	21,32531	-0,53001	23,72088	-0,53001	21,842	-0,53001	21,87958	-0,53001	21,99231
7	-0,54002	-8,49255	-0,54002	14,18556	-0,54002	17,8118	-0,54002	18,33789	-0,54002	19,03307	-0,54002	19,41824	-0,54002	20,51739	-0,54002	19,29612	-0,54002	19,36188	-0,54002	19,70007
8	-0,55003	-8,76498	-0,55003	11,87453	-0,55003	15,72624	-0,55003	15,97049	-0,55003	17,11661	-0,55003	17,46421	-0,55003	18,42243	-0,55003	17,43602	-0,55003	17,40784	-0,55003	17,70846
9	-0,56004	-9,67436	-0,56004	10,34324	-0,56004	13,62189	-0,56004	14,42981	-0,56004	14,82438	-0,56004	15,89534	-0,56004	16,41203	-0,56004	15,49138	-0,56004	15,65109	-0,56004	15,99868
10	-0,57004	-10,3385	-0,57004	9,018632	-0,57004	11,92151	-0,57004	12,95489	-0,57004	13,34946	-0,57004	14,4486	-0,57004	14,35466	-0,57004	14,09161	-0,57004	14,34526	-0,57004	14,2795
11	-0,58005	-11,1296	-0,58005	7,556862	-0,58005	10,26809	-0,58005	11,47057	-0,58005	11,68664	-0,58005	12,40062	-0,58005	12,92671	-0,58005	13,13338	-0,58005	12,50396	-0,58005	12,73882
12	-0,59006	-11,6068	-0,59006	5,683617	-0,59006	9,05621	-0,59006	9,920496	-0,59006	10,44658	-0,59006	10,84115	-0,59006	11,13238	-0,59006	11,03843	-0,59006	11,35784	-0,59006	11,50815
13	-0,60007	-12,1442	-0,60007	4,515892	-0,60007	7,672413	-0,60007	8,50194	-0,60007	9,385015	-0,60007	9,497747	-0,60007	10,48416	-0,60007	10,14596	-0,60007	9,61048	-0,60007	10,409
14	-0,61007	-12,6768	-0,61007	3,210069	-0,61007	6,085698	-0,61007	7,315426	-0,61007	7,873453	-0,61007	8,191924	-0,61007	9,103182	-0,61007	8,774378	-0,61007	8,97166	-0,61007	9,084393
15	-0,62008	-13,8351	-0,62008	2,340147	-0,62008	5,014735	-0,62008	6,697274	-0,62008	7,035472	-0,62008	7,471373	-0,62008	7,461039	-0,62008	7,967398	-0,62008	7,772933	-0,62008	7,626381
16	-0,63009	-14,412	-0,63009	1,44486	-0,63009	4,174875	-0,63009	5,239262	-0,63009	5,933508	-0,63009	5,882778	-0,63009	6,694455	-0,63009	6,394773	-0,63009	6,526295	-0,63009	6,864494
17	-0,6401	-14,085	-0,6401	0,143734	-0,6401	3,235434	-0,6401	4,665263	-0,6401	4,725387	-0,6401	4,948974	-0,6401	5,721194	-0,6401	5,939145	-0,6401	5,540822	-0,6401	6,851777
18	-0,65011	-15,3392	-0,65011	-0,62567	-0,65011	1,646839	-0,65011	3,412988	-0,65011	3,490022	-0,65011	4,076233	-0,65011	4,907639	-0,65011	5,013796	-0,65011	4,666203	-0,65011	1,931125
19	-0,66011	-15,911	-0,66011	-1,9531	-0,66011	1,05781	-0,66011	2,326055	-0,66011	2,567491	-0,66011	3,079487	-0,66011	3,505993	-0,66011	3,464658	-0,66011	4,068718	-0,66011	1,152328
20	-0,67011	-16,000	-0,67011	-0,00754	-0,67011	0,05473	-0,67011	1,000000	-0,67011	0,000000	-0,67011	0,000000	-0,67011	0,000000	-0,67011	0,000000	-0,67011	0,000000	-0,67011	0,000000

Scan
ke-

Jumlah
data

Name

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Today

2

4

6

8

10

20

40

60

80

100

200

400

600

800

1000

Label
Cadmium

Date modified	Type
24/04/2026 14:25	File folder
24/04/2026 13:36	File folder
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